

KUNAL ROY CHOUDHURY

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EXPERIENCE

Software Engineer

March 2026 – Present

CollectEdge

Bengaluru, Karnataka

- Scaled **event-driven microservices architecture** in Rust to **50K+ daily messages** via **horizontal partitioning** across **10+ AWS SQS queues**, with backpressure-aware consumers, circuit-breaker controls, and idempotent processing for AI voice bots, async transcription workers, and WhatsApp campaigns.
- Architected an **AI-powered call analytics platform** in React Native with **polyglot persistence (S3 + PostgreSQL)** and **CQRS** read/write separation, delivering async audio pipelines with ML inference for sentiment/disposition extraction and automatic interaction synthesis.
- Built a **self-optimizing predictive dialer** using weighted fair queueing, **5-tier priority strata**, graph-based contact scoring to maximize agent utilization through real-time routing and ML-assisted dial timing.
- Implemented a **real-time credit risk engine** with **18 parallel rule workers** on reactive streams, multi-bureau aggregation and **40% cost optimization** via LRU cache plus request batching, achieving **sub-100ms** risk scoring.

Software Developer

April 2025 – Feb 2026

Indian Kanoon

Bengaluru, Karnataka

- Architected a **production-grade legal platform** scaling to **1.5 million users** from scratch, leveraging **Django**, **PostgreSQL**, and **Redis** for high availability. Integrated **LLM workflows** across **10+ tools** (including RAG chat & 80+ templates), optimizing system performance for massive concurrent traffic.
- Engineered an **LLM-powered RAG system** leveraging **PostgreSQL text search vectors** with **query expansion algorithms** and context extraction, achieving **15% improvement** in legal search relevance, **92% query accuracy**, and **40% reduction** in latency through **Redis caching** and database optimization.
- Developed **scalable OCR pipelines** using **Tesseract** with **batch processing** for PDF, DOCX, and TXT formats via **Celery async processing**, processing **500+ legal documents** monthly.

Software Engineer Intern

Sept 2024 – April 2025

Chargebee

Chennai, Tamil Nadu

- Architected a **zero-downtime data migration framework** leveraging **JOOQ schedulers** and an **adaptive Vue.js validation system** migrating **500K+ subscriptions** while reducing query latency.

Software Engineer Intern

Jan 2024 – Sept 2024

AiDash

Bengaluru, Karnataka

- Architected a **Spring Boot microservices framework** using **Hexagonal Architecture** to decompose a **2M+ LOC monolith** (accelerating rollouts by **60%**), and optimized SQL pagination for retrieval of **100K+ records**.

PROJECTS

Gostman | *Go, React.js, Wails, REST APIs, JSON*

[link](#)

- Developed a **high-performance desktop API client** using **Wails (Go + React)**, achieving **<1s launch times** and 90% lower memory usage compared to Electron-based alternatives like Postman.
- Engineered a **privacy-first architecture** with local-only storage, supporting **REST, GraphQL, and WebSockets** in a unified interface, complete with **request chaining** for complex workflow automation.

TECHNICAL SKILLS

Generative AI/LLM: RAG systems, LangChain, vector embeddings, OCR (Tesseract)

Programming Languages: Python, Java, Golang, Rust, Dart, JavaScript, TypeScript, SQL

Backend & APIs: Django, Spring Boot, Express.js, RESTful APIs, GraphQL, async task processing (Celery)

Frontend & Mobile: React, React Native, Vue.js, Flutter, Redux, Zustand, Tailwind

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, PostgreSQL text search, ORMs

Cloud & DevOps: AWS (EC2, S3, SQS, RDS, CloudFront), Docker, Kubernetes, CI/CD, Git, GitHub Actions

EDUCATION

SRM Institute of Science and Technology

Kattankulathur, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.24

Aug. 2020 – Sept 2024